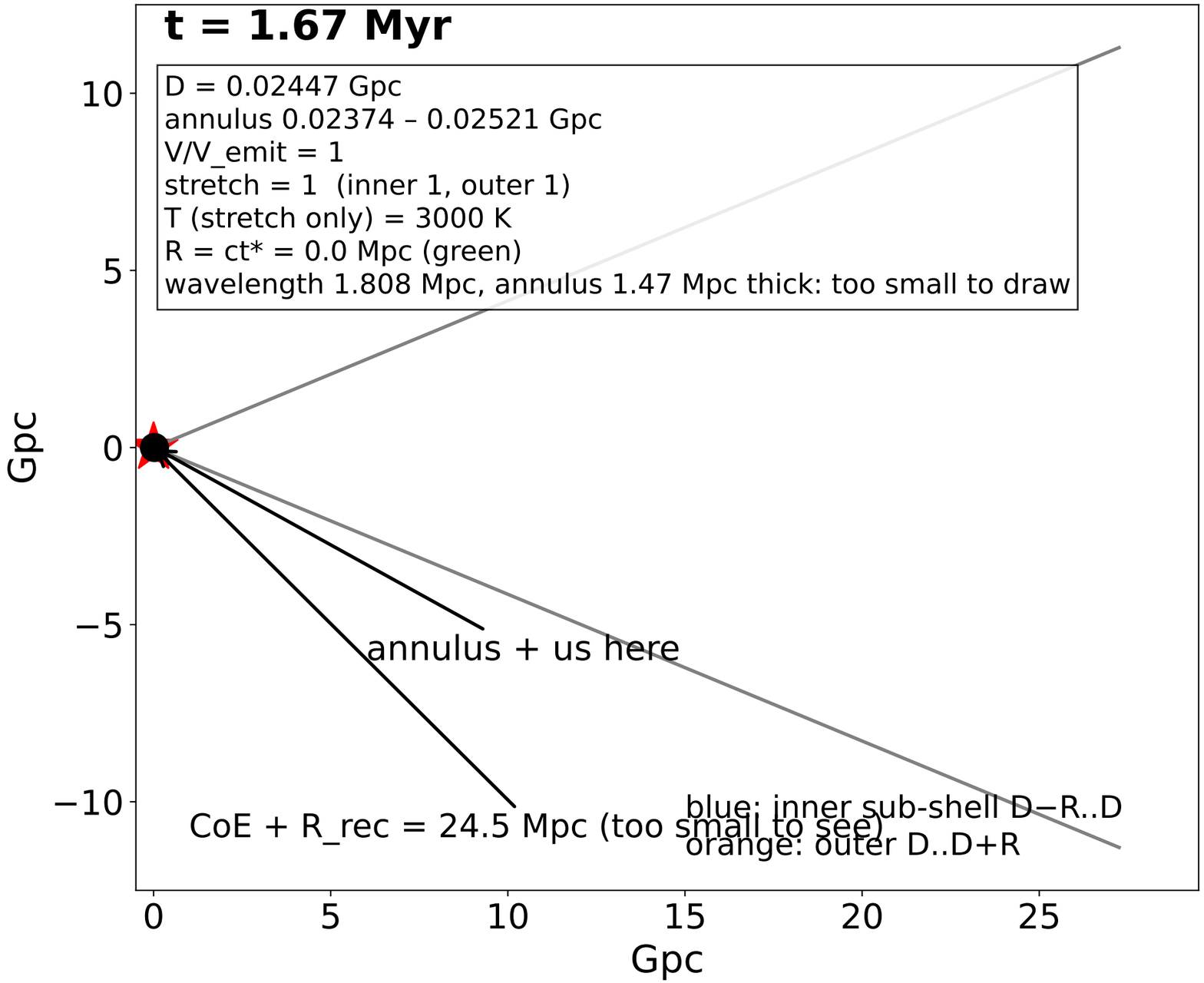
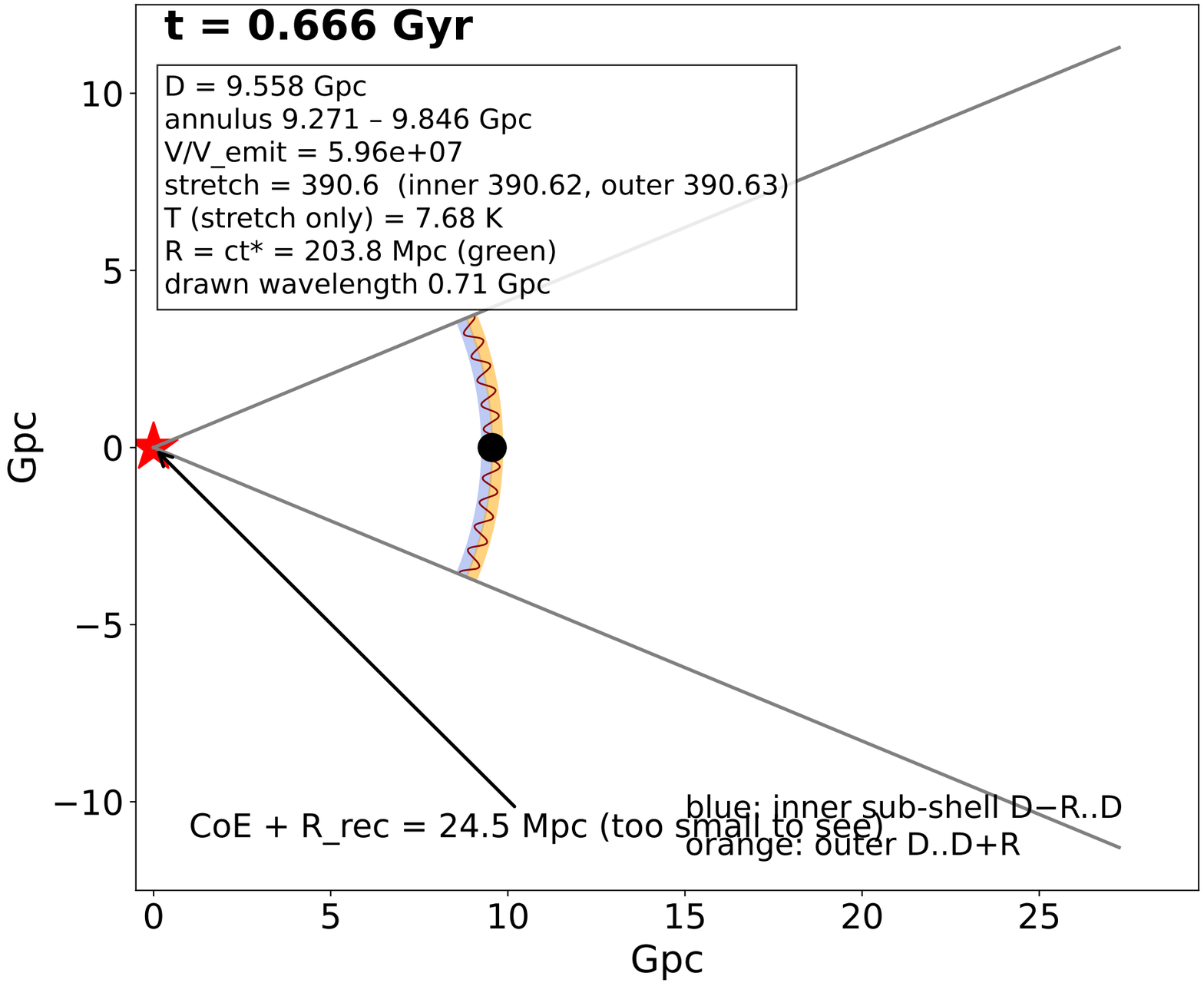
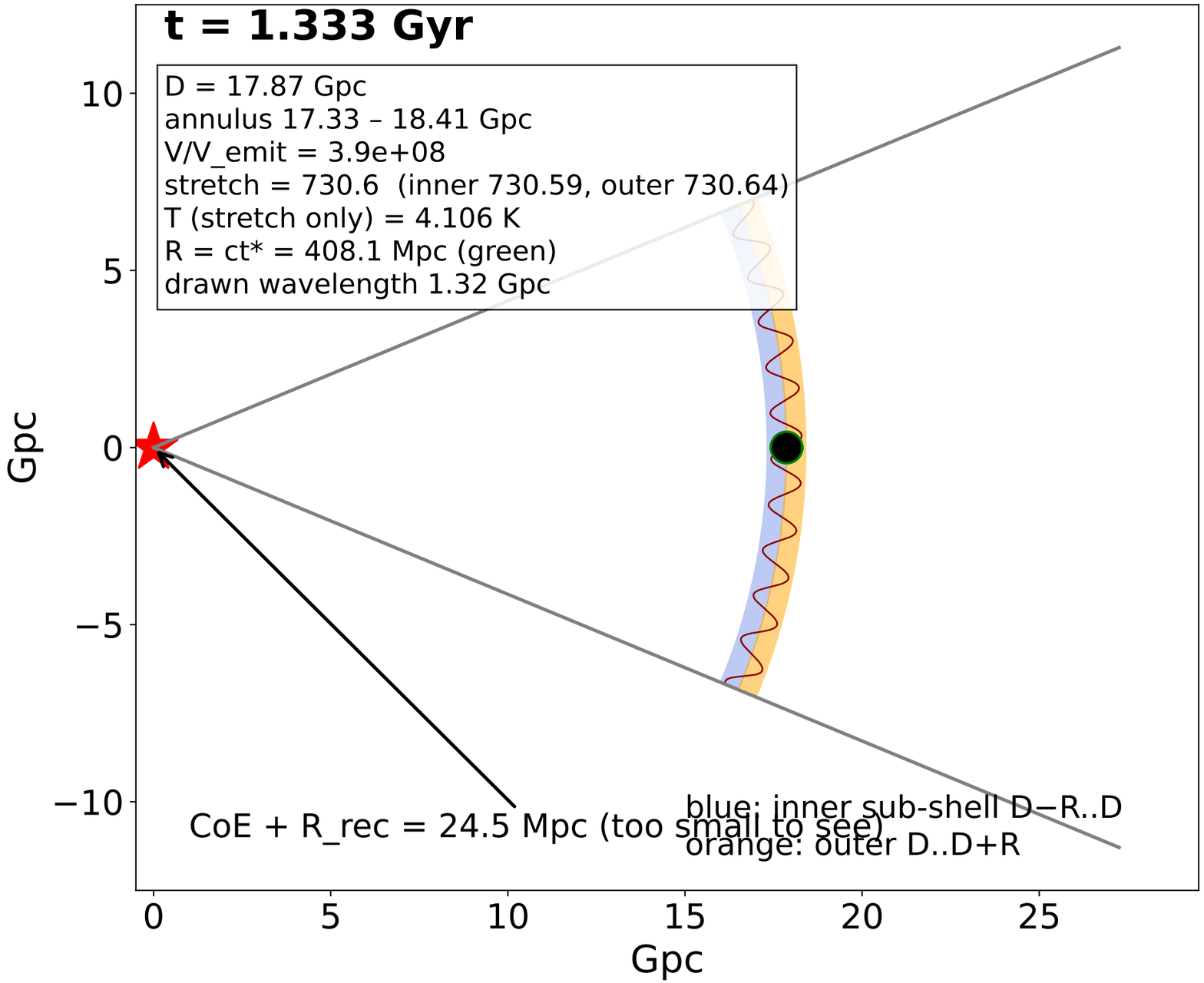
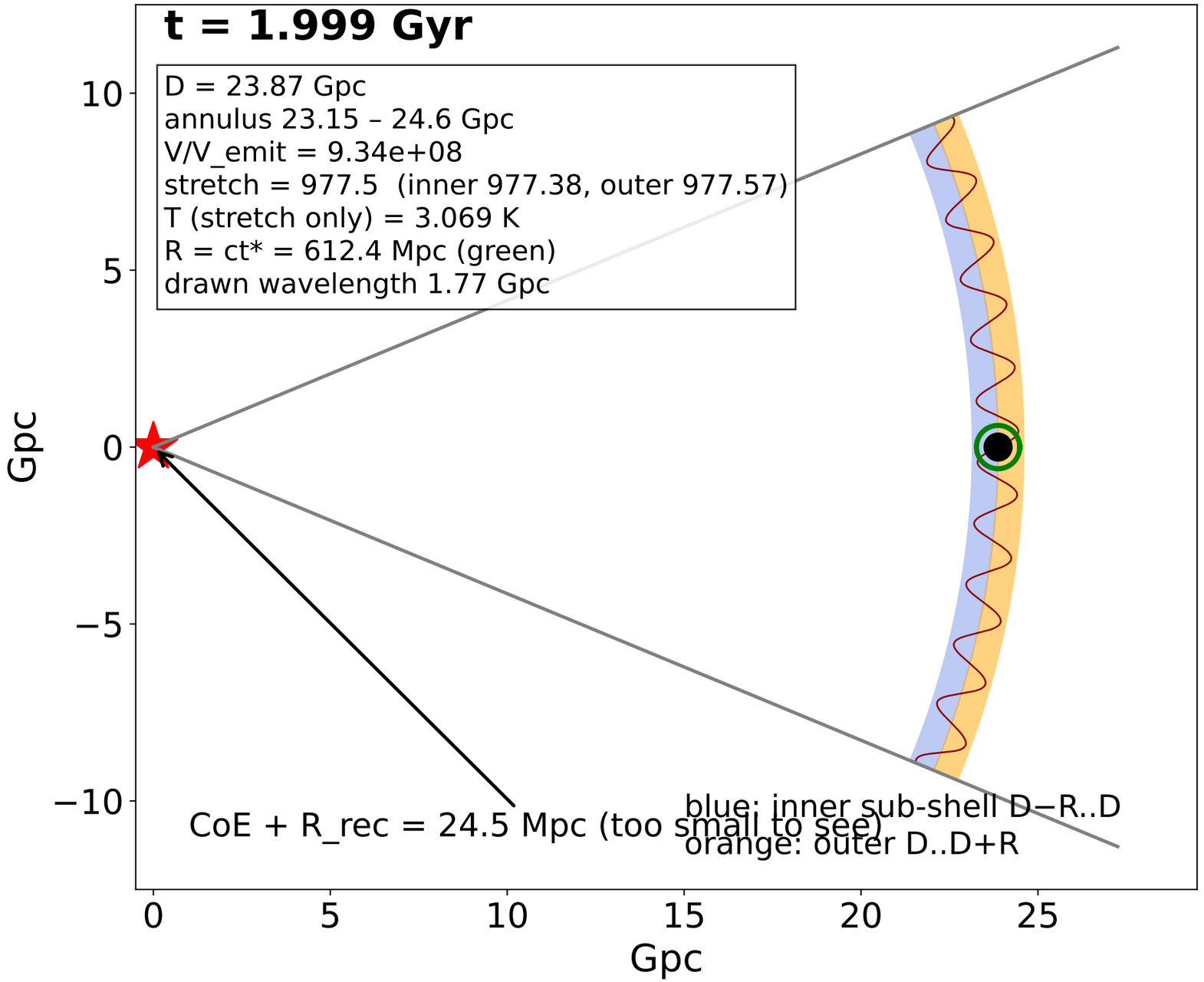
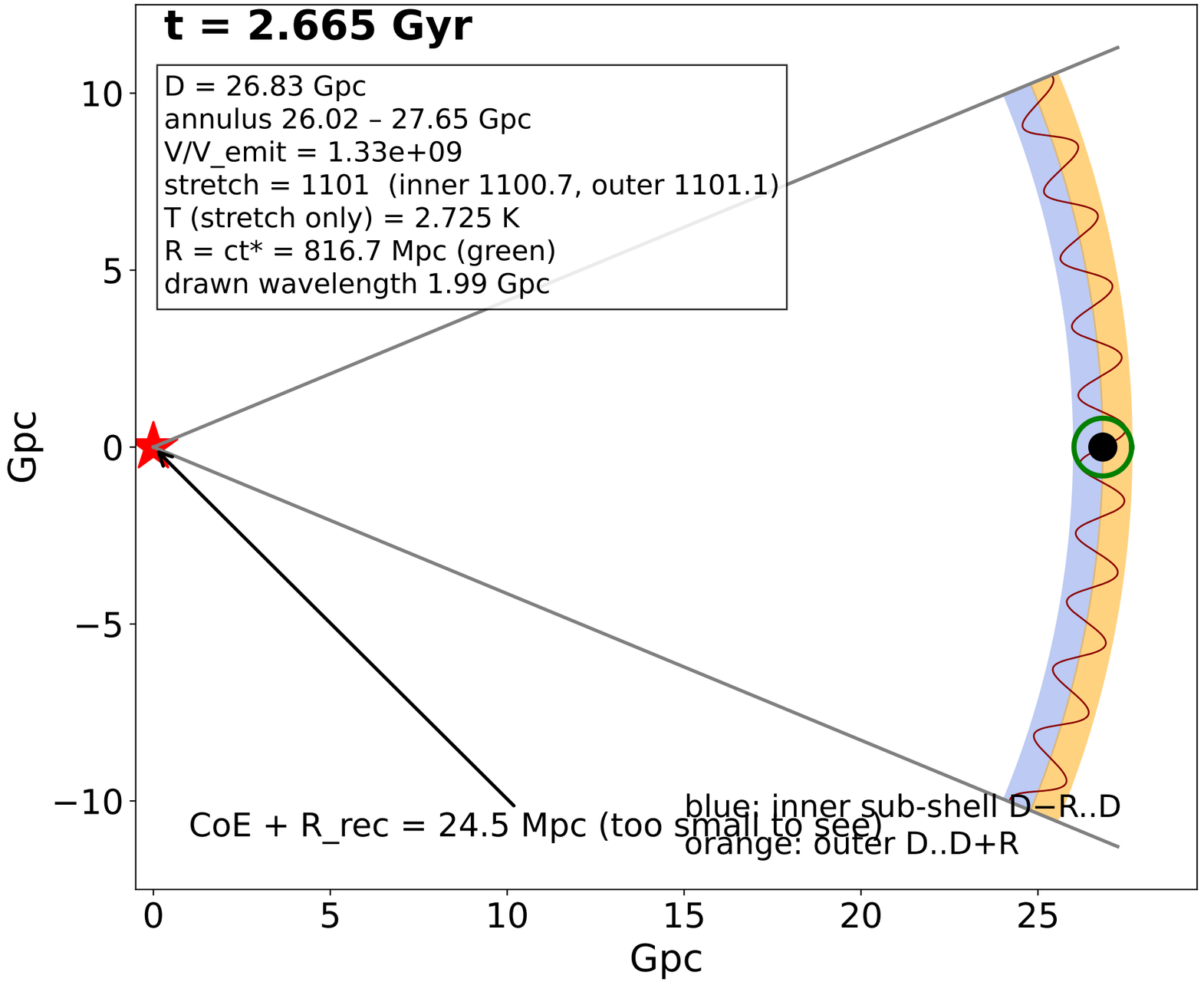
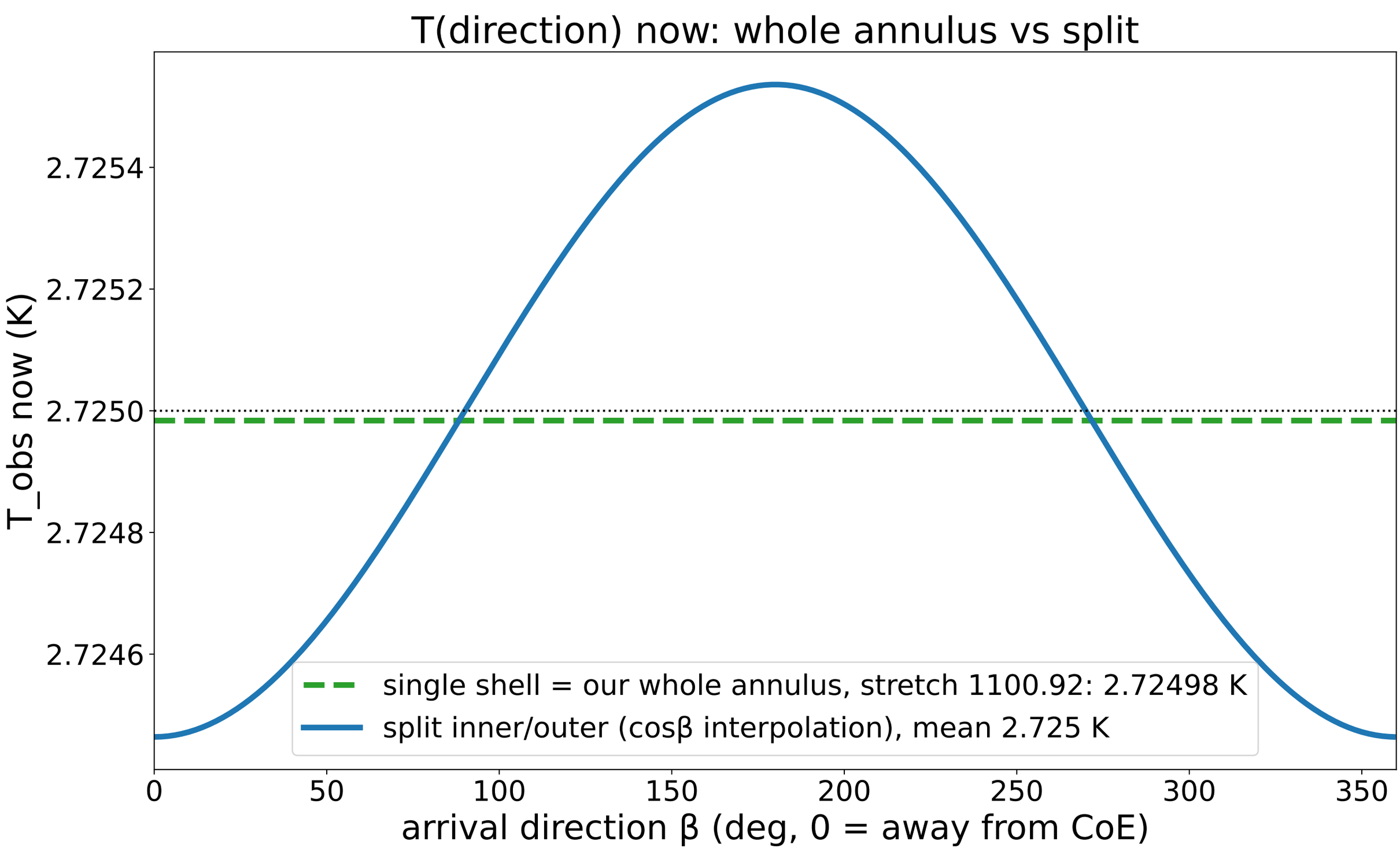

t = 1.67 Myr
D = 0.02447 Gpc
annulus 0.02374 – 0.02521 Gpc
V/V_emit = 1
stretch = 1 (inner 1, outer 1)
T (stretch only) = 3000 K
R = ct* = 0.0 Mpc (green)
wavelength 1.808 Mpc, annulus 1.47 Mpc thick: too small to draw
annulus + us here
CoE + R_rec = 24.5 Mpc (too small to see)
blue: inner sub-shell D−R..D
orange: outer D..D+R
Gpc
Gpc
10
5
0
−5
−10
0
5
10
15
20
25

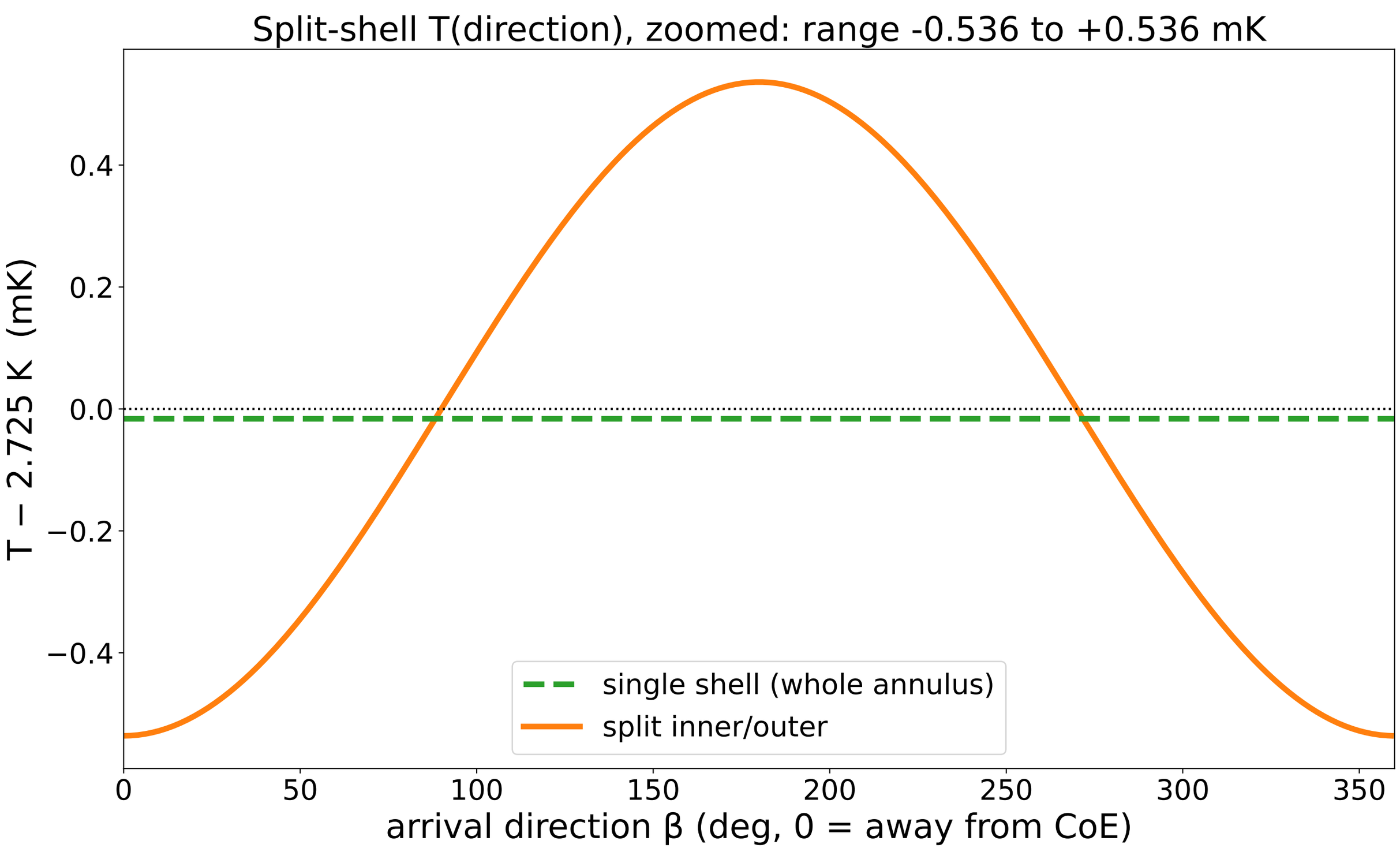

t = 0.666 Gyr
D = 9.558 Gpc
annulus 9.271 – 9.846 Gpc
V/V_emit = 5.96e+07
stretch = 390.6 (inner 390.62, outer 390.63)
T (stretch only) = 7.68 K
R = ct* = 203.8 Mpc (green)
drawn wavelength 0.71 Gpc
CoE + R_rec = 24.5 Mpc (too small to see)
blue: inner sub-shell D−R..D
orange: outer D..D+R
Gpc
Gpc
10
5
0
−5
−10
0
5
10
15
20
25

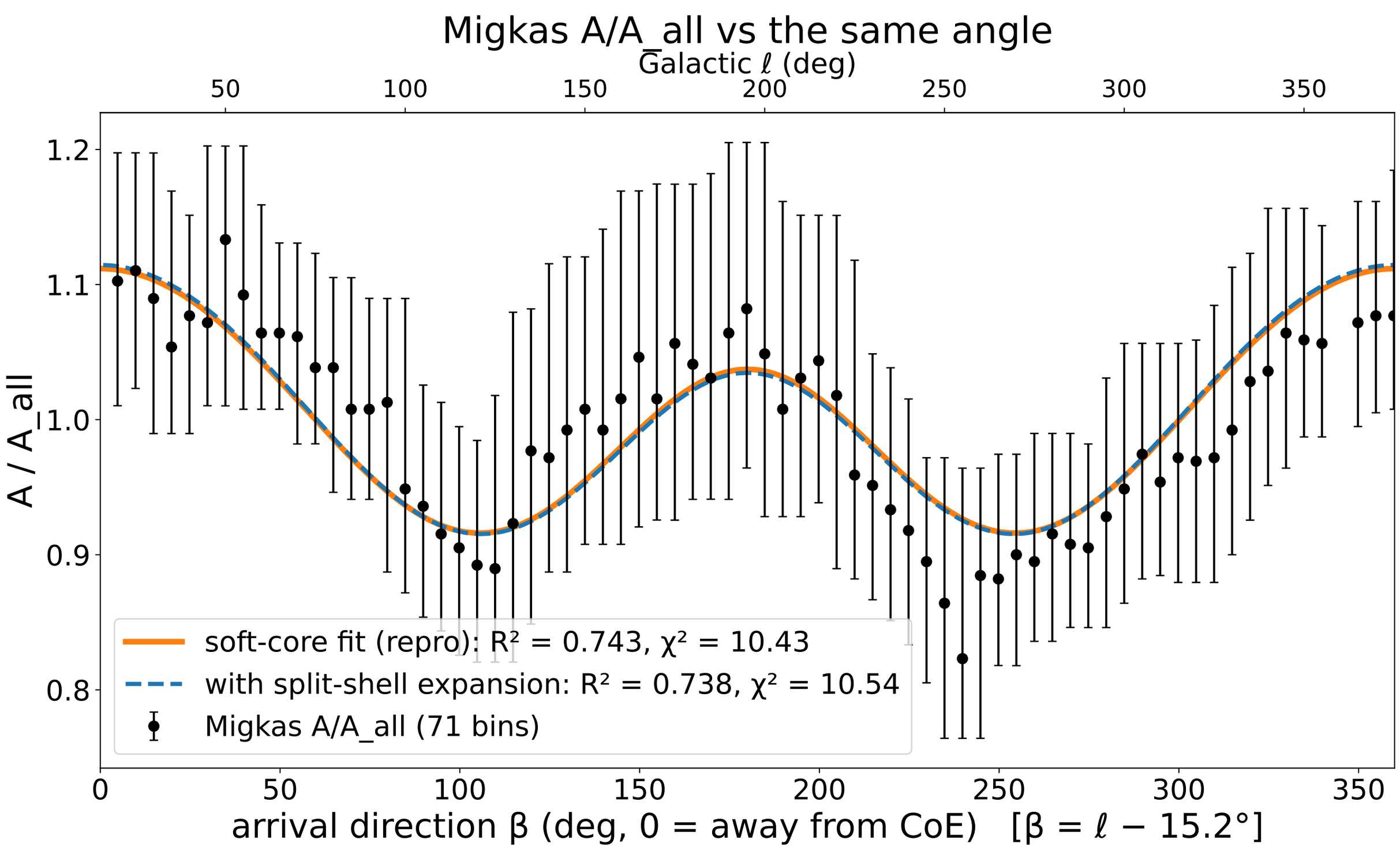

t = 1.333 Gyr
D = 17.87 Gpc
annulus 17.33 – 18.41 Gpc
V/V_emit = 3.9e+08
stretch = 730.6 (inner 730.59, outer 730.64)
T (stretch only) = 4.106 K
R = ct* = 408.1 Mpc (green)
drawn wavelength 1.32 Gpc
CoE + R_rec = 24.5 Mpc (too small to see)
blue: inner sub-shell D−R..D
orange: outer D..D+R
Gpc
Gpc
10
5
0
−5
−10
0
5
10
15
20
25

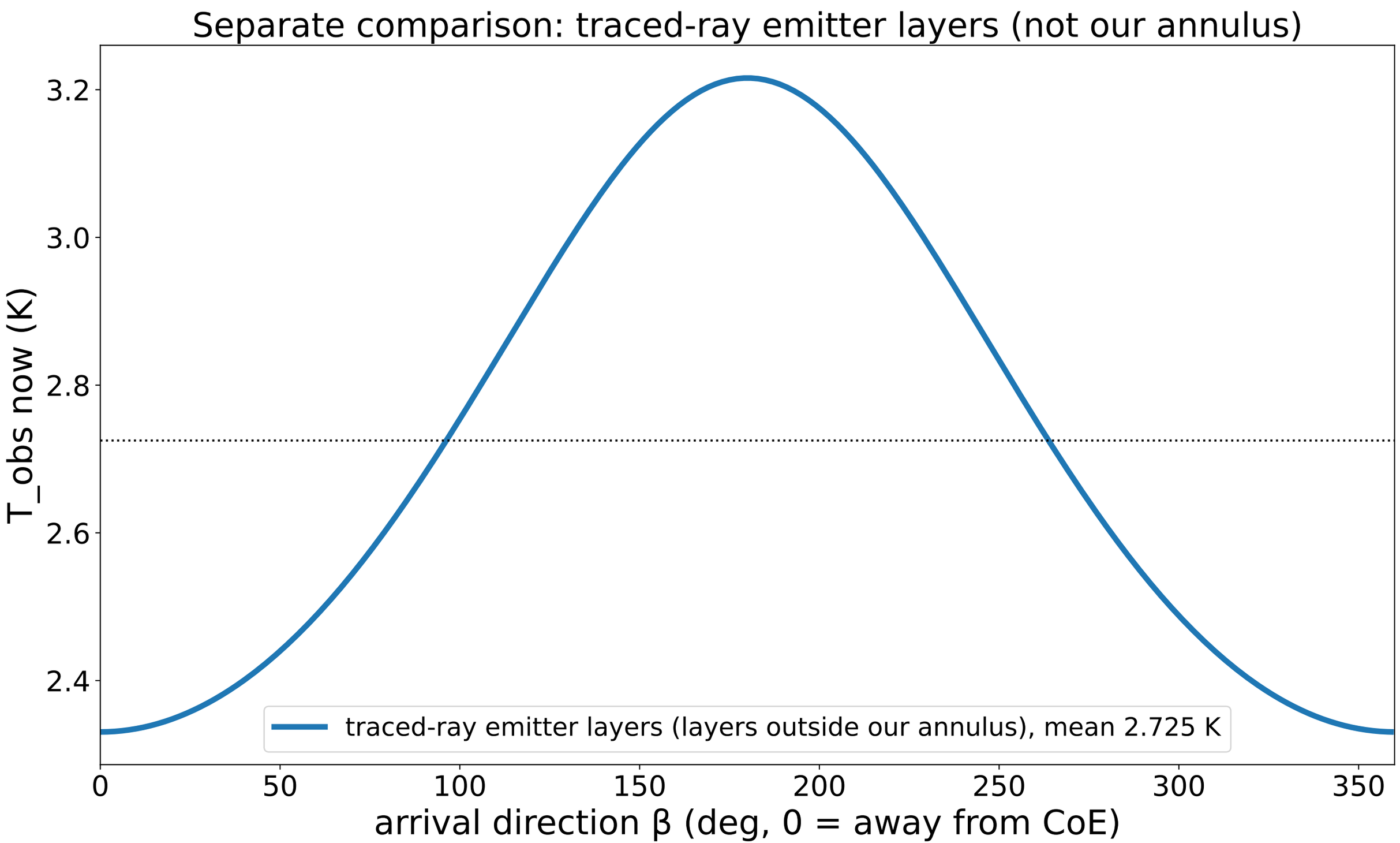

t = 1.999 Gyr
D = 23.87 Gpc
annulus 23.15 – 24.6 Gpc
V/V_emit = 9.34e+08
stretch = 977.5 (inner 977.38, outer 977.57)
T (stretch only) = 3.069 K
R = ct* = 612.4 Mpc (green)
drawn wavelength 1.77 Gpc
CoE + R_rec = 24.5 Mpc (too small to see)
blue: inner sub-shell D−R..D
orange: outer D..D+R
Gpc
Gpc
10
5
0
−5
−10
0
5
10
15
20
25


t = 2.665 Gyr
D = 26.83 Gpc
annulus 26.02 – 27.65 Gpc
V/V_emit = 1.33e+09
stretch = 1101 (inner 1100.7, outer 1101.1)
T (stretch only) = 2.725 K
R = ct* = 816.7 Mpc (green)
drawn wavelength 1.99 Gpc
CoE + R_rec = 24.5 Mpc (too small to see)
blue: inner sub-shell D−R..D
orange: outer D..D+R
Gpc
Gpc
10
5
0
−5
−10
0
5
10
15
20
25


T(direction) now: whole annulus vs split
T_obs now (K)
2.7254
2.7252
2.7250
2.7248
2.7246
0
50
100
150
200
250
300
350
arrival direction β (deg, 0 = away from CoE)
single shell = our whole annulus, stretch 1100.92: 2.72498 K
split inner/outer (cosβ interpolation), mean 2.725 K


Split-shell T(direction), zoomed: range -0.536 to +0.536 mK
T − 2.725 K (mK)
0.4
0.2
0.0
−0.2
−0.4
0
50
100
150
200
250
300
350
arrival direction β (deg, 0 = away from CoE)
single shell (whole annulus)
split inner/outer


Migkas A/A_all vs the same angle
Galactic ℓ (deg)
50
100
150
200
250
300
350
1.2
1.1
1.0
0.9
0.8
A / A_all
soft-core fit (repro): R² = 0.743, χ² = 10.43
with split-shell expansion: R² = 0.738, χ² = 10.54
Migkas A/A_all (71 bins)
0
50
100
150
200
250
300
350
arrival direction β (deg, 0 = away from CoE) [β = ℓ − 15.2°]


Separate comparison: traced-ray emitter layers (not our annulus)
T_obs now (K)
3.2
3.0
2.8
2.6
2.4
0
50
100
150
200
250
300
350
arrival direction β (deg, 0 = away from CoE)
traced-ray emitter layers (layers outside our annulus), mean 2.725 K